blood

723. CLINICAL ALLOGENEIC AND AUTOLOGOUS TRANSPLANTATION: LATE COMPLICATIONS AND APPROACHES TO DISEASE RECURRENCE: POSTER III | NOVEMBER 29, 2018

# A Novel Unifrac-Based Analytical Method for Determining Sequential Changes in Gut Microbiota after Hematopoietic Stem Cell Transplantation

Shinsuke Kusakabe, MD,[*,1] Kentaro Fukushima, MD PhD,[*,1] Tetsuo Maeda, MD,[*,1] Takafumi Yokota, MD PhD,[1] Ritsuko Nakai, MD,[*,1] Akihisa Hino, MD,[*,1] Yasuhiro Nagate, MD,[*,1] Jiro Fujita, MD PhD,[*,1] Motooka Daisuke,[*,2] Shota Nakamura,[*,2] Hirohiko Shibayama, MD PhD,[1] Yuzuru Kanakura, MD PhD[3]

[1]Department of Hematology and Oncology, Osaka University Graduate School of Medicine, Suita, Japan

[2]Resarch Institute of Microbial Diseases, Osaka University, Suita, Japan

[3]Department of Hematology and Oncology, Osaka University Graduate School of Medicine, Suita, Osaka, Japan

***Blood*** **(2018) 132 (Suppl_1) : 4589.**

**http://doi.org/10.1182/blood-2018-99-116064**

## Abstract

The human gut harbors various microorganisms, which are important for maintenance of systemic health, including metabolism and immune system homeostasis. Recent studies demonstrated the association of intestinal flora with various diseases such as inflammatory bowel disease and rheumatoid arthritis. In addition, studies on patients with hematopoietic stem cell transplantation (HSCT) showed that the variation in intestinal bacterial flora after transplantation is related to prognosis, frequency of infection, sensitivity to antimicrobial agents, and severity of graft versus host disease. Nonetheless, most studies showing less diversity of intestinal flora evaluated patients' feces a certain period after HSCT. In order to determine the relationship between the alterations in intestinal flora composition and the clinical course post-HSCT, we used metagenomic analysis to evaluate the time-dependent changes in intestinal bacterial flora in patients undergoing allogeneic (allo) HSCT, and compared them with those of healthy controls (HC) or recipients of autologous (auto) peripheral blood HSCT.

Patients who received autologous or allogeneic HSCT at the Osaka University Hospital were enrolled after obtaining informed consent. We collected 613 feces samples from 24 patients from August 2016 to May 2017 before and after transplantation. Samples were collected in sterile centrifuge tubes and 16S

rRNA deep sequencing was performed using MiSeq (Illumina). In total, 543 specimens were available for analysis.

First, we evaluated the diversity of the microbiota in pre-transplantation specimens using Simpson's and Shannon-Wiener index. However, neither method revealed any significant difference between auto/allo-patients and HC despite intensive treatment history of the patients, but showed that the major phyla of gut microbiota contained *Bacteroides*, Firmicutes, and *Fusobacteria* in all three groups. Next, we utilized 'uniFrac distance' to evaluate both the diversity in community structure and members. Indeed, this method was useful for evaluating both weighted (quantitative) and unweighted (qualitative) variations of numerous organisms. Using this method, we were able to accurately estimate the abundance of gut microbiota. Interestingly, weighted uniFrac distance, which describes distances in community structure, was significantly wider between allo vs HC and HC vs HC ($p < 0.001$) and auto vs HC and HC vs HC ($p < 0.001$). Furthermore, unweighted uniFrac, which describes the distance between community members, was significantly wider between allo vs HC and HC vs HC ($p < 0.001$), but not between auto vs HC and HC vs HC ($p = 0.69$).

Next, we analyzed the structural components of microbiota using pre-treatment samples. Results showed that the population of beneficial bacteria such as butyric acid-producing bacteria, *Coprococcus*, *Faecalibacterium*, and *Lachnospiraceae*, were significantly reduced in the microflora of pre-transplantation patient samples compared to HC. In addition, sequential chronological analyses revealed that dynamic fluctuation in the intestinal flora was induced in patients by use of antibiotics. We also observed inter-patient variation in the extent of fluctuation; > 50% patients lost flora diversity after transplantation, whereas some patients retained it despite administration of broad spectrum antibiotics.

Next, we analyzed whether alteration in flora composition and/or the loss of diversity affected patient survival. Among 24 patients, 16 showed altered flora composition, whereas 8 retained the flora characteristics. Among these 16 cases, the 2-year mortality of patients who had lost flora diversity was 63.6 %, whereas it was 25.0 % for patients who retained the diversity ($p < 0.001$). In contrast, all 8 patients who retained original flora composition were alive. Hence, we concluded that both alteration in flora composition and loss of diversity are strongly associated with patient survival.

In conclusion, commonly used diversity indexes do not completely describe the differences within genera of gut microbiota, and uniFrac analysis is a powerful tool for evaluating gut microbiota in patients with HSCT, as it analyzes community composition/structure in detail. This method can also be used to simultaneously evaluate alterations in flora composition and loss of diversity, both of which affect survival post-HSCT.

**Disclosures**

**Yokota:** *Pfizer Inc.:* Research Funding; *Bristol-Myers Squibb:* Research Funding; *MSD K.K.:* Research Funding; *CHUGAI PHARMACEUTICAL CO., LTD.:* Research Funding; *Celgene:* Research Funding. **Shibayama:** *Ono Pharmaceutical Co.,LTD:* Honoraria, Research Funding; *Celgene K.K.:* Honoraria, Research Funding; *Bristol-Meyer Squibb K.K:* Honoraria, Research Funding; *Mundipharma K.K.:* Honoraria, Research Funding; *Fujimoto Pharmaceutical:* Honoraria, Research Funding; *Jansen Pharmaceutical K.K:* Honoraria; *Novartis Pharma K.K.:* Honoraria, Research Funding; *Takeda Pharmaceutical Co.,LTD.:* Honoraria, Research Funding. **Kanakura:** *Alexion Pharmaceuticals, Inc.:* Consultancy, Honoraria, Research Funding.

# Author notes

* Asterisk with author names denotes non-ASH members.